

Taxing Lost Comparison: Decision Information, Private Testing, and Costed Corrective Instruments

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Abstract

Can the loss of informative comparison create an economically meaningful corrective-tax margin? This paper derives that margin from a declared decision experiment rather than treating transaction counts as monetary damage. A Poisson number of independent Gaussian trials informs a binary adoption decision. The resulting value of evidence is increasing, strictly concave, and bounded. A firm jointly chooses productive or restrictive activity and private testing, while activity reduces an organic stream of informative trials. The firm captures only a share of the continuation benefit. A globally corrective payment schedule therefore has two margins: activity and evidence production. An activity-only charge cannot implement an interior first best with positive private testing and incomplete internalization. Public testing can displace private trials one for one; it creates additional information only after the private target is exceeded. With a fixed setup cost and different unit costs, the efficient laboratory decision has an exact threshold, and available replacement testing caps the marginal cost of lost organic comparison. Numerical examples are stipulated and reproducible, not empirical tax estimates. The comparison-budget result supplies a conditional statistical constraint; welfare valuation, causation, appropriation, and instrument feasibility remain separate premises.

Keywords: corrective taxation, information externalities, comparison budget, experimentation, public evaluation, second-best policy.

JEL codes: H21, H23, D62, D83, L41.

1. Introduction

Technological activity can simultaneously produce useful services and change the evidence available about alternatives. An exclusive route, compatibility condition, or dominant allocation process may reduce opportunities for independent trials. If those trials inform consequential future choices and their benefit is not fully appropriated, the change can create an information externality. The fiscal question concerns the marginal net external effect of an identified choice. Firm size, profitability, or a statistical impossibility theorem cannot by themselves answer it. The distinction is especially important when private evaluation responds. An incumbent that loses organic feedback may buy substitute tests. A government funding public evaluation may then displace the incumbent's own spending rather than increase information. An activity tax intended to preserve comparison may leave the private testing margin distorted. An alternative intervention could replenish evidence more cheaply, while a fixed laboratory cost can make small programs unattractive. These interactions are absent when the value of information is simply assumed to decline linearly with activity.

This paper develops a decision experiment and a policy game that solve these margins together. An unknown binary state determines the preferred continuation action. A Poisson count of independent Gaussian trials generates evidence. The money value is the reduction in the specified decision's expected loss, with its prior and payoffs held fixed. A firm chooses activity and private trials. Activity has private production benefits, other external benefits or damage, and a negative effect on an organic trial stream. The firm captures a fraction of the common continuation value, and public trials may be purchased at a different cost.

The main results are conditional but exact. Evidence value has positive, strictly decreasing marginal returns. A nonlinear schedule aligning the full private objective with welfare contains

both an activity charge and a testing subsidy. An activity-only instrument cannot implement an interior two-choice optimum when the firm internalizes less than the full information benefit.

Private testing responds to public provision through a target total intensity, giving exact crowdout. A fixed-cost laboratory is chosen only if its cost advantage generates sufficient optimized surplus. Replacement testing also caps the marginal welfare cost of losing an organic trial.

These results separate statistical scarcity from fiscal correction. The comparison budget in the supplied October 2026 Shadow Futures manuscript bounds separation in a declared allocation experiment. Under its further hypotheses, a finite budget gives equivalent complete-history laws and prevents universally consistent inference from one history. That fact does not identify a causal effect of activity, an external monetary loss, or a tax base. The associated public record is DOI 10.2139/ssrn.6003994; its July revision is distinct from the local October version used here. The present paper introduces the additional economic structure needed for those steps.

The model can apply to procurement evaluation, platform testing, or a shared technological assessment service when its assumptions are credible. It does not estimate a tax on current AI companies. Known logged assignment probabilities are not automatically an unknown structural recipient kernel, and reward labels can provide additional information. A field application must specify the full observation law and the decision affected by it. Productive gains, privacy costs, innovation responses, and public financing constraints then belong in the same welfare calculation.

The contribution sought is therefore an explicit decision-and-instrument bridge. Corrective taxation, information externalities, and endogenous learning have substantial prior literatures. The global implementation identity is a classical benchmark. The proposed model-specific

additions concern joint testing incentives, crowding out, fixed-cost provision, and the replacement-cost envelope within a common calibrated experiment. Their wider priority requires further review. Keeping these boundaries clear allows a strong conditional result without claiming an identified fiscal remedy from information scarcity alone.

2. Literature and the distinction from a generic Pigouvian argument

Arrow (1962) explains the economic relevance of uncertainty, knowledge production, and incomplete appropriation. Coase (1960) makes institutions, transaction costs, and feasible alternatives central to externality analysis. Greenwald and Stiglitz (1986) give a broad framework for externalities under imperfect information and incomplete markets. The quasilinear model here is a narrower benchmark with a specified prior, an observable activity, and public likelihoods. It does not derive a general failure of markets from the mere existence of uncertainty.

Blackwell (1953) orders experiments by their value for decision problems. Bergemann and Morris (2019) survey information design in games, and Bergemann and Bonatti (2019) examine information markets and appropriation. Jones and Tonetti (2020) emphasize nonrival data and incentives to restrict its use. Our experiment concerns producing and substituting informative trials, which precedes any distribution of their record. The benefit is derived for a fixed continuation decision; a scalar trial count has value only through the declared observation law and loss function.

The option to ignore a signal is not an unconditional welfare theorem for strategic disclosure. Morris and Shin (2002) show that public information can have ambiguous social effects with coordination motives. Accordingly, the continuation decision here is a single common objective whose payoff is divided among beneficiaries. There is no game in which publication itself

changes strategic actions in a socially costly way. Such effects can be added as external terms, but they cannot be dismissed by citing the monotonicity of a private decision value.

Endogenous learning is established economic theory. Rothschild (1974) analyzes market pricing as experimentation, and Brezzi and Lai (2000) study incomplete learning from endogenously generated data. The comparison-budget framework adds a conditional way to control allocation-history separation, not the first observation that choices affect learning. In the present game, organic and purchased trials are interchangeable informative experiments. This assumption enables exact crowdout; if private tests and independent public tests reveal different dimensions, substitutability and policy conclusions must be reconsidered.

Sandmo (1975) analyzes optimal taxation with externalities, while Bovenberg and de Mooij (1994) show the importance of preexisting fiscal distortions. The globally corrective schedule below uses nondistortionary transfers and a fixed operating firm. It is therefore a first-best benchmark, rather than an assertion that environmental or other corrective rates transfer unchanged to information. Saez and Stantcheva (2016) make social welfare weights explicit; the sum-of-surplus objective here is one transparent choice, not a distribution-free measure of social desirability. Farhi and Gabaix (2020) further show how behavioral departures alter optimal taxation; our firm is fully rational under its stated prior.

Weitzman (1974) compares prices and quantities under uncertainty. Kaplow and Shavell (2002) emphasize how allowing nonlinear harm schedules changes that comparison. The present paper accordingly distinguishes a whole payment schedule from a marginal rate at one point and from an activity-only instrument. Barthold (1994) organizes practical tax-design questions about the base, payer, rate, and ancillary constraints. Here those issues translate into verifying informative

trial intensity, measuring private appropriation, and preventing a proxy tax from rewarding uninformative exposure.

Manski (2013) motivates retaining policy uncertainty rather than substituting an unsupported point estimate. That is especially relevant when the state separation, causal access effect, or decision payoff is uncertain. A model-specific first best is a useful benchmark for comparing instruments, not evidence that a real tax authority knows the required schedule. The article's additional content lies in the explicit Poisson value, the impossibility of correcting both private choices with one activity instrument, and the costed comparison of public and private evidence provision. These claims can be evaluated independently of any assertion that existing tax systems undercorrect a particular sector.

3. A consequential evidence experiment

Fix a binary structural state $\theta \in \{-1, +1\}$ with equal prior probability. A continuation decision chooses one of two technologies or interventions. Selecting the state-appropriate action pays $L > 0$, while selecting the wrong action pays zero. Without evidence, either action succeeds with probability one half, so expected payoff is $L/2$. The payoff and action set are held fixed across policies. This is an explicit bounded decision problem, not a social value assigned to statistical precision without identifying what the precision is for.

Let N have a Poisson distribution with intensity $\lambda \geq 0$, independent of θ . Conditional on $N=n$, observe n independent measurements X_j distributed normally with mean $\theta\mu$ and variance σ^2 , where $\mu, \sigma > 0$ are publicly known. All trial sources share this observation law and have conditionally independent shocks. Counts are observable and state-independent. The informative object is the measurement, not the fact that an allocation occurred. Define $d = \mu/\sigma$ and let Φ be the standard normal distribution function.

For $n \geq 1$, the likelihood-ratio decision chooses +1 when the sample sum is positive. Its success probability under either state is $\Phi(d\sqrt{n})$. At $n=0$, either tie decision has success probability one half. Hence the exact incremental value of the experiment is

$$V(\lambda) = L \left[e^{-\lambda} \sum_{n=0}^{\infty} \frac{\lambda^n \Phi(d\sqrt{n})}{n!} - \frac{1}{2} \right]. \quad (1)$$

The series converges absolutely because its probability-weighted terms are bounded. It evaluates a Bayesian decision under a stipulated likelihood. Calibration would require evidence for that likelihood and for L . Multiple beneficiaries can share the decision payoff; their shares do not change the aggregate value. A different prior, asymmetric losses, correlated trials, or a different action set changes (1). Nothing in the comparison-budget theorem supplies μ , σ , or L automatically.

Proposition 1. $V(0)=0$, V is strictly increasing and strictly concave on $[0, \infty)$, and V approaches $L/2$ as λ tends to infinity. Writing $f(n)=\Phi(d\sqrt{n})$, its derivatives are

$$\begin{aligned} V'(\lambda) &= L\mathbb{E}[f(N+1) - f(N)] > 0, \\ V''(\lambda) &= L\mathbb{E}[f(N+2) - 2f(N+1) + f(N)] < 0. \end{aligned} \quad (2)$$

In particular $V'(0)=L[\Phi(d)-1/2]$. Every finite λ leaves strictly positive expected classification error. The joint likelihood divergence is $\text{KL}(P_+^\lambda \| P_-^\lambda) = 2d^2\lambda$, and incremental value satisfies

$$V(\lambda) \leq \frac{L}{2} \min\{1, d\sqrt{\lambda}\}. \quad (3)$$

Proof. For $x > 0$, $f'(x) = d\phi(d\sqrt{x})/(2\sqrt{x}) > 0$ and its logarithmic derivative is $-1/(2x) - d^2/2 < 0$. Thus f extends continuously as a strictly concave function at zero, so its forward differences are positive and strictly decreasing. Differentiating the bounded Poisson transform gives (2). As λ increases without bound, N tends to infinity in probability; bounded convergence yields the limit. At finite λ , the event $N=0$ alone contributes positive error. Conditional normal-measurement KL is $2d^2$ per trial; averaging over the common count law gives $2d^2\lambda$. The equal-prior testing identity and Pinsker's inequality prove (3). Appendix A justifies the transform differentiation.

The family is ordered by informative trial intensity: an experiment at a larger λ can be thinned independently to reproduce a smaller λ , so it is Blackwell more informative. This ordering comes from the trial law and independent thinning, not from a count alone. If added exposures carry no state-dependent measurement, their information value is zero. If they merely repeat perfectly correlated evidence, this family is the wrong model. The precise assumptions are also why V has diminishing returns here; arbitrary economically important experiments need not have the same curvature.

4. Endogenous activity and private appropriation

A firm chooses activity $a \in [0, A]$ and private test intensity $e \geq 0$. Activity can combine a productive operating scale with a specified access condition. Its private operating surplus is $\Pi_0(a)$, continuously differentiable. Let $G(a)$ and $D(a)$ be additional noninformational benefits and harm to others. An organic stream supplies intensity $b(a) > 0$, continuously differentiable and decreasing. Public testing supplies $r \geq 0$. Independent Poisson streams add, so the evidence intensity is $\lambda = b(a) + e + r$. Their conditional measurements have the common law in Section 3. Privately financed measurements are delivered to the same continuation decision maker as organic and public trials.

Private tests cost $c_e > 0$ per unit of expected intensity. Public tests cost $c_r > 0$ per unit plus a real setup expense $K_r \geq 0$ whenever $r > 0$. Setup is a resource expense, not a tax receipt. The firm captures share $\omega \in [0, 1]$ of the aggregate continuation benefit, with the remaining share accruing to external decision beneficiaries. Discounting is $\delta \in (0, 1]$. Write $I_{(r>0)}$ for the indicator that the public facility is active. The private and social objectives are:

$$\Pi(a, e; r) = \Pi_0(a) - c_e e + \omega \delta V(b(a) + e + r). \quad (4)$$

$$W(a, e, r) = \Pi_0(a) + G(a) - D(a) - c_e e - c_r r - K_r I_{(r>0)} + \delta V(b(a) + e + r). \quad (5)$$

All terms share the decision horizon, numeraire, and resource boundary. Operating benefits in Π_0 and G exclude the continuation information benefit in V . Customer payments and firm receipts have opposite signs in aggregate surplus; a displaced profit is not automatically a resource loss. Harm D includes any real privacy or quality cost omitted from production, but such costs are not assumed positive because a firm is large. The scalar activity need not be production quantity, and its unit must be declared before a rate can be interpreted.

The discount and payoff split give a tractable internalization mechanism. External beneficiaries may be future consumers, other firms, or a procurement authority unable to contract for all later uses. Complete contracting would move ω toward one. If the firm values a different decision, or private testing reveals proprietary signals unavailable to external beneficiaries, one common V no longer suffices. That alternative requires a vector of experiments and payoffs. The split in (4) and (5) is therefore a substantive hypothesis, not an empirical claim that a particular percentage of AI learning benefits is external.

For fixed a and r , strict concavity of V gives a unique optimal private e . The bounded value and positive testing costs prevent unbounded testing expenditure. Compact a and continuity ensure a maximizer in (a,e) after restricting e to a sufficiently large harmless bound; a larger e cannot offset its linear cost with $V \leq L/2$. Joint concavity in a and e is not automatic because b can be convex. The global implementation result below does not require it. Uniqueness or a stated selection rule is needed when government implementation depends on which maximizer the firm chooses.

5. The globally corrective schedule and a missing-instrument result

Hold public provision r fixed for the moment. The external payoff relevant to the firm's joint choices is $E(a,e;r)=G(a)-D(a)+(1-\omega)\delta V(b(a)+e+r)$. Public resource costs are constants with

respect to those choices. Suppose activity and informative private intensity can be verified, nonlinear positive and negative payments are feasible, and revenue is transferred nondistortionarily. These strong conditions define a benchmark against which constrained instruments can be assessed.

Proposition 2. The total payment schedule

$$T(a, e; r) = D(a) - G(a) - (1 - \omega)\delta V(b(a) + e + r) + K \quad (6)$$

implements every social maximizer over (a, e) for the fixed r , subject to the same feasible set.

This is global objective alignment, with no concavity requirement. Its two marginal instruments are

$$\begin{aligned} T_a &= D' - G' - (1 - \omega)\delta V'(\lambda)b'(a), \\ T_e &= -(1 - \omega)\delta V'(\lambda). \end{aligned} \quad (7)$$

Proof. Subtracting (6) from (4) gives $\Pi - T = W + c_r r + K_r I_{(r>0)} - K$. The remaining terms are

constant in (a, e) , so the maximizing sets coincide. Differentiation gives (7). If government commits to r , uses this schedule, and can select or uniquely induce the relevant conditional maximizer, maximizing the induced welfare over r also selects a global first best. The discrete setup decision must be evaluated by objective values, not differentiated as if it were smooth.

Because $b' < 0$, the information part of T_a is positive whenever $\omega < 1$. Noninformational benefits G enter with the opposite sign. Meanwhile T_e is a subsidy to an informative measurement margin.

These rates vary with total evidence: diminishing marginal value reduces both as substitutes accumulate. The lump-sum K does not affect conditional activity choices, but matters for operating participation, entry, budget balance, and distribution. An implementation treating the firm as fixed cannot declare K harmless when it changes whether the firm operates.

The schedule is the classical identity $T = -E + K$ applied to an explicitly derived decision value. Its substantive content here lies in the two-choice structure. A rate equaling (7) at one target only aligns local first-order conditions. It may leave other global maxima, especially with nonconvex

b or a fixed facility cost. A uniform proportional tax on fully deductible true economic profit instead leaves the maximizing choices unchanged in this benchmark. Real corporate-tax systems have other bases and financing effects, so that observation is not a prediction for every actual tax system.

Proposition 3. Fix r and suppose a first best has interior a^* and $e^*>0$, $c_e>0$, and $\omega<1$. No differentiable payment depending only on a can implement that same pair as a private optimum. The conclusion permits boundary exceptions and does not rule out welfare-improving activity taxation.

Proof. The social first-order condition for e is $\delta V'(\lambda^*)=c_e$. An activity-only payment leaves the private e condition $\omega\delta V'(\lambda^*)=c_e$. Their difference requires $(1-\omega)c_e=0$, contrary to the premises. The contradiction is independent of the curvature in a . An activity instrument can move a and indirectly e , but cannot make the two incompatible marginal conditions hold at the specified interior pair.

This result explains why preserving an organic trial stream is not equivalent to correcting evidence production. At an interior private optimum the firm replaces part of any lost stream using its own tests. Yet its target information remains below the social target. A levy on activity alone may correct other external effects while leaving that target unchanged. A testing subsidy, evidence-purchase contract, or public provision may be required. Verifying e means verifying an informative trial under the specified law; subsidizing an easily counted but uninformative allocation does not implement (6).

6. Private testing, public crowdout, and the constrained activity margin

Define λ_ω by $\omega\delta V'(\lambda_\omega)=c_e$ when $\omega\delta V'(0)>c_e$. Otherwise set $\lambda_\omega=0$. Strictly decreasing V' and its zero limit ensure a unique positive solution in the first case. If $\omega=0$, private testing is never

worthwhile and the zero definition applies. λ_ω is the firm's desired total informative intensity, not a target number of suppliers or an equilibrium market concentration.

Proposition 4. For each fixed a and r , the unique private test response is

$$e_p(a, r) = \max\{0, \lambda_\omega - b(a) - r\}. \quad (8)$$

While $e_p > 0$, an extra unit of public intensity displaces one unit of private intensity, leaving total evidence exactly λ_ω . Once e_p reaches zero, additional public tests increase total intensity. On an interior private-testing branch, a marginal reduction in organic intensity is also replaced one for one by private spending.

Proof. Differentiating (4) in e gives $\omega \delta V'(b+e+r) - c_e$. It is strictly decreasing in e . An interior zero requires total intensity λ_ω , while a nonpositive derivative at $e=0$ makes zero optimal. The derivative of (8) is -1 in r and $-b'(a)$ in a on the positive branch. These relations hold away from the kink and follow directly from the common trial law and linear cost. Different signal quality, capacities, or private information can invalidate the one-for-one response.

Crowdout need not make public provision worthless. At fixed a on this branch, changing r by a small amount changes real resource cost by $c_r - c_e$ while leaving information value constant.

Public tests can save resources if $c_r < c_e$. If $c_r \geq c_e$ and there are no other external effects or capacity limits, they waste or merely relabel resources before crossing the private target. The setup cost can eliminate a small apparent saving. A comparison must therefore count displaced private spending as well as the purchased public stream.

If activity remains within the positive-private-testing branch, the firm's optimized payoff is $\Pi_0(a) + c_e b(a) + c_e r - c_e \lambda_\omega + \omega \delta V(\lambda_\omega)$. A public-intensity change adds a constant to this branch objective, so it does not change a unique interior branch optimum. It can nevertheless change the global optimum when the activity choice crosses a kink. The conditional result should not be

read as invariance of activity under arbitrary public programs. For exponential b , sufficient operating curvature can secure a unique branch solution; Appendix B states a simple condition. There is a second-best implication when only activity can be taxed. Along the positive private response, total evidence is constant. Welfare after substituting e_p has derivative $\Pi_0' + G' - D' + c_e b'$. The firm's optimized derivative is $\Pi_0' + c_e b'$. Thus the activity correction on this branch is $D' - G'$, not the partial information term in (7). The firm already pays the replacement cost of losing organic trials, while the remaining information undersupply is constant along this response. Applying the two-instrument partial derivative as an activity-only formula would ignore that substitution.

When $e_p = 0$, the compensated response is unavailable and a change in b changes total information. The constrained activity correction then includes $-(1-\omega)\delta V'b'$ again. At the kink, one-sided derivatives and global values govern implementation. This branch analysis is a model-specific application of second-best reasoning: which margin is constrained changes the relevant behavioral derivative. It is not a claim that information externalities generally disappear whenever private evaluation exists.

7. Costed public evaluation and replacement value

Now solve the planner's evidence-purchase problem at a fixed activity. For a unit cost $c > 0$, let λ_c be zero if $\delta V'(0) \leq c$ and otherwise the unique solution to $\delta V'(\lambda_c) = c$. Define the optimized gross continuation surplus after buying additional intensity y by

$$F_c(b) = \max_{y \geq 0} \{\delta V(b + y) - cy\}, \quad y_c(b) = \max\{0, \lambda_c - b\}. \quad (9)$$

With perfectly substitutable independent trials, unlimited private capacity, and no distinct publicity benefit, the cheaper active source supplies all purchased intensity. If $c_r \geq c_e$ and $K_r \geq 0$, public provision is weakly dominated by private testing at the same intensity. When $c_r < c_e$,

compare using the public facility with the private route, incorporating its real setup cost. This comparison concerns optimally corrected provision, not the private uncorrected response in Section 6.

Proposition 5. At a fixed b with $c_r < c_e$, define $H(b) = F_{(c_r)}(b) - F_{(c_e)}(b)$. H is nonnegative, continuous, and nonincreasing. A positive public facility is strictly preferred to the private route exactly when $H(b) > K_r$. Where both routes buy positive intensity, $H'(b) = c_r - c_e < 0$. Where only the public route buys, $H'(b) = c_r - \delta V'(b) < 0$. Once $b \geq \lambda_{(c_r)}$, $H(b) = 0$.

Proof. Any purchase feasible at c_e costs weakly less at c_r , so optimized gross surplus is no smaller. Formula (9) follows from strict concavity. Its envelope derivative is c when $b < \lambda_c$ and $\delta V'(b)$ when $b \geq \lambda_c$, with the derivatives agreeing at the boundary. Since $\lambda_{(c_r)} > \lambda_{(c_e)}$ whenever their positive targets exist, subtraction gives the stated regions and signs. The branch where neither purchases has zero difference. Subtracting the setup cost yields the facility decision. If $K_r = 0$ and $H = 0$, an inactive public facility and no extra testing tie; the strict preference statement concerns $H > K_r$.

The facility is valuable where missing organic evidence makes replacement sufficiently costly and its unit advantage covers setup. A large nominal market does not by itself imply that threshold is crossed. Qualification standards, independence, and evidence dissemination may change c_r or the signal law. Public and private trials that are not substitutes require a different allocation problem. Likewise, a capacity-constrained incumbent can make public provision useful even when its nominal unit cost is higher. Those are meaningful extensions, not exceptions to the stated theorem.

Corollary. Within a chosen smooth evidence route, the marginal value of one additional organic trial after optimal replacement is

$$F_c'(b) = \min\{\delta V'(b), c\}. \quad (10)$$

Consequently the marginal welfare cost of the organic loss caused by activity is

$-b'(a)\min\{\delta V'(b(a)), c\}$. When replacement is active, it equals the cost of replacing the trial rather than its unmitigated decision loss. Proof follows directly from the envelope calculation. This cost cap relies on a feasible informative substitute with unlimited capacity. It disappears when the proposed replacement is uninformative or technically infeasible.

Joint activity choice requires a global comparison of routes. Define $Q(a) = \Pi_0(a) + G(a) - D(a)$. The private-testing route maximizes $Q(a) + F_{(c_e)}(b(a))$; the public route maximizes

$Q(a) + F_{(c_r)}(b(a)) - K_r$, allowing an inactive route if appropriate. Compare the optimized values, not merely H at the uncorrected private activity. A cheaper route can change the efficient a because it lowers the resource cost of replacing lost organic evidence. Facility adoption also creates nonconvexity through K_r . First-order equations describe a smooth active branch but do not decide which branch should exist.

The result gives a disciplined alternative to taxing a presumed irreversible information loss. If replacement is affordable, the relevant social sacrifice may be its cost. If it is costly or unavailable, preservation of the organic experiment can matter more. A charge, access remedy, and evaluation program can therefore be substitutes or complements depending on capacity, internalization, and setup cost. Their ranking must come from this comparison rather than an assumption that public testing always adds evidence or that every lost comparison has an arbitrarily large shadow price.

8. A reproducible stipulated example

Set $L=10$, $d=1$, $\delta=0.95$, $\omega=0.4$, $c_e=0.6$, $c_r=0.3$, and $K_r=0.3$. Let $a \in [0,4]$, $\Pi_0(a)=2a-a^2/2$, $G(a)=0.2a$, $D(a)=0.1a^2$, and $b(a)=2\exp(-a/2)$. The positive operating returns and external benefit

preserve a reason for useful activity. The declining organic stream is a stipulated causal mechanism, not an estimate from a company. Trial and activity costs use arbitrary common monetary units.

Evaluation of the Poisson series gives the private target $\lambda_{\omega}=1.04613$. The social target using private tests is $\lambda_{(c_e)}=2.41325$; using public tests it is $\lambda_{(c_r)}=3.55699$. The uncorrected private choice is $a=1.74987$, organic intensity 0.83378, and private intensity 0.21235. Total social surplus at that allocation, with no public facility and the normalization in (5), is 4.23811. Thus positive private evaluation does not eliminate the gap between private and social evidence targets.

Correcting both choices and using private tests yields $a=1.60976$, organic intensity 0.89428, purchased intensity 1.51896, and welfare 4.75996. The public route yields $a=1.72796$, organic intensity 0.84296, public intensity 2.71403, zero private tests, and welfare 5.07297 after setup expense. The cheaper replacement route permits more activity than the corrected private route while providing more information. It does not maximize activity or prescribe a sector-wide rate. At the uncorrected activity, adding public intensity 0.2 reduces private intensity to 0.01235 and leaves total intensity 1.04613. It adds no decision evidence over that range, although cheaper provision saves resources before setup is counted. The optimized public-route advantage at this fixed activity is $H=0.62323$, exceeding the setup cost 0.3. The joint route decision still uses the separately optimized values above. All figures are solutions to the given model, rounded to five decimals.

For reproduction, compute $f(n)=\Phi(\sqrt{n})$, sum the Poisson transform and its forward-difference derivative, and solve the three strictly monotone target equations by bisection. Substitute the exact target response into each one-dimensional objective and compare endpoints with stationary

points. For $\lambda \leq 4$, truncating after $n=50$ changes the bounded-value series by less than $\text{LPr}\{\text{Poisson}(4) > 50\}$; this is below 10^{-35} . The displayed results were also checked with 200 terms. Numerical illustration adds no empirical evidence about external harm or actual taxation.

9. Proxy instruments, financing, and uncertainty

Suppose the authority can tax only a proxy $z(a,e)$ through a differentiable payment $g(z)$. At a proposed interior allocation, local correction of both margins requires $g'(z)\nabla z = -\nabla E$. The two gradient vectors must be collinear. If the same proxy value arises at two allocations requiring different proportionality factors, a single derivative g' cannot supply both. A flat proxy cannot correct a nonzero external gradient. This is an elementary rank restriction, not a proof that every proxy tax is useless.

For example, a proxy depending on activity alone has zero derivative in e and fails under Proposition 3. Compute usage that counts production and evaluation together may likewise mix margins with opposite corrective signs. A policy rewarding trial counts must distinguish independent informative measurements from state-independent allocations or redundant repetitions. Administrative observability and economic relevance are separate requirements. A direct evidence-purchase contract can act on the correct margin only when its quality and delivery are enforceable.

The benchmark cancels taxes and subsidies as transfers and treats setup expense as real cost. Distortionary public financing changes that calculation. A subsidy can require revenue with an excess burden, and a tax can interact with labor supply, investment, or another tax base. Sandmo (1975) and Bovenberg and de Mooij (1994) show why those interactions need an actual second-best system. Multiplying every rate here by an assumed marginal cost of funds is not a derived

solution. A restricted government budget must enter the policy optimization and the firm's response together.

Within this model one transparent restriction is an administratively fixed public budget for tests:

$c_r r + K_r I_{(r>0)} \leq B_g$. It bounds feasible public intensity and can prevent the least-cost target from being reached. The private route remains available, so capacity and budget bind differently from a ban on private evaluation. A full corrective schedule might also be fiscally infeasible. One should then maximize W over implementable schedules and induced choices, not claim that the unconstrained target survives unchanged. Participation, tax avoidance, and cross-border relocation require further constraints.

Model uncertainty is equally consequential. The information margin depends on state separation d , decision payoff L , appropriation ω , and the causal derivative b' . An observed association between activity and fewer trials does not identify b' . If admissible models yield different signs for $G'-D'$ or different trial quality, a single positive corrective rate may not be identified. Manski (2013) supports retaining the feasible model set and comparing decision losses. A robust choice requires an explicit criterion such as expected welfare under a declared prior or minimax regret; it is not entailed by the impossibility theorem.

A welfare gain under the sum-of-surplus objective is not automatically a Pareto improvement. Some beneficiaries, taxpayers, or incumbent owners may lose. Actual compensation requires transfers that are feasible and consistent with the budget. Social welfare weights can represent a different distributional objective, following the concern formalized by Saez and Stantcheva (2016). They change both decision valuation and instrument choice. Nor does the fixed operating-firm model decide optimal industry entry, productive investment, or jurisdictional tax competition. Those extensions must state their additional agents and feasible actions.

10. Comparison budgets and an application protocol

The comparison budget concerns allocation probability outside the current leader. In a bounded conditional-logit experiment it controls conditional Fisher information and finite-horizon likelihood separation. Ordered-pair domination and an almost-surely finite budget can yield equivalent complete-history laws. That conclusion concerns one specified experiment and a fixed candidate cohort. It neither makes any scalar budget a money value nor implies that every possible independent evaluation law is also equivalent. New experiments can reveal parameters the original adaptive record cannot identify.

For a binary consequential decision, finite-horizon separation bounds can limit attainable value through total variation, as in (3). The chain rule requires expected cumulative budget when bounding expected KL. Almost-sure finite budget alone does not guarantee a finite expected budget and does not supply a numerical damage estimate. Infinite budget also need not identify a parameter if the observations do not vary with it. These distinctions prevent an information bound from being converted into a fiscal charge by substituting a raw count for λ or V .

A proposed application should specify the decision, beneficiaries, and errors; identify the structural states and full observed outcomes; estimate the quality and independence of informative trials; measure how a feasible activity change affects their arrival; and determine which benefits private actors appropriate. Only then can trial costs and implementation constraints be compared. A shared evaluation stream may be appropriate when public tests reveal usable evidence at lower cost. Where private responses displace it or its measurements do not discriminate the relevant states, a different design is needed.

The distinction is concrete in machine-learning feedback. A known logging policy may allocate exposures identically under competing structural states, while clicks or preference labels contain

information. Those responses enter the joint observation law; the logging probability is not automatically p_β from the allocation theorem. Conversely, a forced-exposure quota without any state-dependent measurement can have zero information value. Public evaluation must protect the informative feedback channel, rather than nominally increasing the count of alternatives. The empirical deliverable would be an estimated or bounded intervention effect on decision loss, plus private and public substitution costs. It should compare the baseline, an activity remedy, an evidence subsidy, a public facility, and feasible combinations. Real productive and coordination gains remain in Π_0 and G . The model supplies falsifiable response predictions and a reproducible welfare calculation. It does not establish that existing AI firms are undertaxed or that an optimal corporate-tax percentage can be read from their scale.

11. Conclusion

Informative comparison has an economic value only relative to a consequential decision and a specified observation law. The Poisson experiment derives that value and permits joint analysis of activity, private evaluation, and public provision. A full corrective schedule contains two margins. An activity-only tax cannot implement an interior first best when private testing remains underinternalized. Public trials can crowd out private trials without adding evidence, while lower costs can still justify provision.

With a costed substitute, the marginal sacrifice from losing organic comparison is capped by its replacement cost. A fixed public setup expense then creates an explicit facility threshold and requires global route comparison. These results make the policy question sharper than a generic Pigouvian analogy: determine which evidence is lost, whose decision suffers, how private behavior responds, and what least-cost instrument can change it. The comparison-budget

framework constrains the original experiment; those economic and institutional premises determine the remedy.

12. Appendix A: Poisson value and separation

Write $p_n(\lambda) = \exp(-\lambda)\lambda^n/n!$. For $\lambda > 0$, $p_n' = p_{(n-1)} - p_n$, with $p_{(-1)} = 0$. On compact λ intervals, bounded f and Poisson tails justify termwise differentiation. Shifting indices gives $dE[f(N)]/d\lambda = E[f(N+1) - f(N)]$, and a second application gives the second difference in (2). At zero, the power-series expansion gives the same right derivatives. Since strict concavity of f yields strictly negative second differences at every integer, $V'' < 0$ including its right limit at zero. The forward differences tend to zero, so $V'(\lambda)$ tends to zero as N grows in probability. Conditional on n , the sample sum has mean $\theta n\mu$ and variance $n\sigma^2$. The likelihood ratio is $\exp(2\mu \sum X_j / \sigma^2)$, producing the stated classifier and success probability. The two Gaussian product laws have positive densities and KL $2nd^2$. Mixing over a common Poisson count adds no count divergence, so joint KL is its expectation $2d^2\lambda$. Equal-prior optimal success probability is $[1+TV]/2$, hence $V = LTV/2$. Applying $TV \leq \sqrt{KL/2}$ and $TV \leq 1$ gives (3). At $\lambda=0$, both laws coincide and all these quantities are zero.

For the monotone experiment ordering, take $\lambda_2 > \lambda_1$. Independently retain each observed measurement with probability λ_1/λ_2 . Poisson thinning gives count $\text{Poisson}(\lambda_1)$, and retained measurements preserve their state-dependent independent normal law. This state-independent randomized map reproduces the smaller experiment. It also proves monotonicity for other bounded decision problems under the same fixed-state family, although the strict concavity of (1) uses the particular symmetric classifier payoff. The proof does not assert concavity for arbitrary information design.

13. Appendix B: Optimization and instrument checks

For $c > 0$, the derivative of $\delta V(b+y) - cy$ is strictly decreasing in y . It crosses zero at $\lambda_c - b$ if that is positive; otherwise the constrained maximum is zero. The resulting F_c is differentiable at $b = \lambda_c$ because $\delta V'(\lambda_c) = c$. This gives the envelope in (10) and all cases of Proposition 5. If $\delta V'(0) \leq c$, no purchased tests are socially worthwhile at any $b \geq 0$ and the same formula reduces to $F_c = \delta V(b)$. A positive facility in the proof is selected only where its saving exceeds setup expense.

The fixed setup cost does not create an existence problem. Compare the inactive branch $r=0$ with the branch allowing nonnegative purchases and charging K_r . Their continuous optimized values attain maxima over compact activity; positive test costs bound all purchases. If the best charged branch occurs at zero purchase, it cannot strictly dominate the inactive branch when $K_r \geq 0$. This construction establishes a maximizer without differentiating the indicator or assuming joint concavity. Equal-cost sources or tied branch values yield multiple optimal implementations.

For $\Pi_0(a) = pa - ca^2/2$ and $b(a) = b_0 \exp(-\gamma a)$, the positive-private-testing branch has second derivative $-c + c_e \gamma^2 b(a)$. The sufficient condition $c > c_e \gamma^2 b_0$ makes it strictly concave. Its first-order condition is $p - ca - c_e \gamma b(a) = 0$, with boundary conditions if necessary. In the example $c=1$ exceeds $0.6 \times 0.25 \times 2 = 0.3$. The corrected private route in its active region has derivative $p + g - (c + d_a)a - c_e \gamma b$, where $G = ga$ and $D = d_a a^2/2$; the public route replaces c_e by c_r . These equations verify the reported stationary points.

The proxy restriction follows from the chain rule $\nabla T = g'(z) \nabla z$. For local alignment, $\nabla T = -\nabla E$. If $\nabla z = 0$ and $\nabla E \neq 0$, no alignment is possible; if both are nonzero they must be parallel with the correct scalar. Two points with the same z impose the same g' , so inconsistent required scalars rule out a common schedule. This checks a local corrective vector. It does not by itself establish global implementation, which requires the objective alignment or a separate incentive proof.

The numerical series uses bounded normal probabilities and Poisson weights. For the reported target range, the omitted value tail after $n=50$ is at most L times a $\text{Poisson}(4)$ tail, and derivative tails have the same probability bound up to a fixed bounded-difference factor. Targets were found by bisection and each active activity equation compared with the interval endpoints and kink. No randomness or empirical data enter these calculations. Rounding affects displayed numbers only, not the symbolic propositions.

14. Statements and declarations

Data availability. The article is theoretical and uses no observational data. The stipulated example is reproducible from equation (1), its derivative in (2), the target equations, and the explicit parameter list. All reported numbers are model calculations, not estimates. No new human-subject or field data were collected.

AI assistance. OpenAI Codex was used for literature retrieval, development of the explicit models, generative drafting, numerical calculation, and algebraic checking. This substantive assistance is disclosed. The human author remains responsible for verifying the model, proofs, references, calculations, and final text before submission.

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